

3.8 Polymerisation

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.8 Polymerisation
Difficulty	Hard

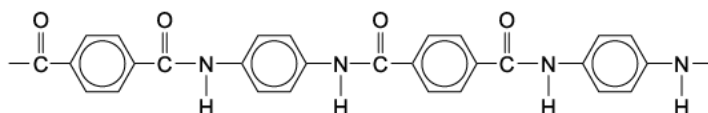
Time allowed: 10

Score: /5

Percentage: /100

Question 1

Kevlar has the structure below.



A Kevlar rope is both lighter and stronger when compared to a steel rope of similar dimensions.

Which properties of Kevlar are responsible for this?

- 1 The Kevlar molecule has no permanent dipole.
- 2 The fibres of Kevlar align due to hydrogen bonding.
- 3 The mass per unit length is less in a Kevlar rope than in a steel rope.

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Question 2

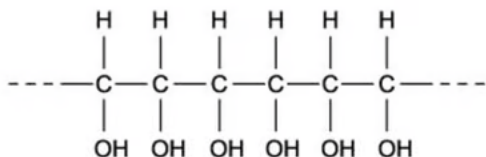
Which row correctly describes what happens when poly(propene), $(C_3H_6)_n$, is burned in an excess of air?

	Moles of O_2	Products
A	Poly(propene) does not burn	
B	$1\frac{1}{2}n$	$3nC + 3nH_2O$
C	$4\frac{1}{2}n$	$3nCO_2 + 3nH_2O$
D	$6n$	$3nCO_2 + 3nH_2O$

[1 mark]

Question 3

The following diagram represents the structure of a polymer.



By which method can this polymer be made?

	Step 1	Step 2
A	Polymerise ethene	Hydration
B	Polymerise 1,2-dichloroethene	Hydrolysis
C	Polymerise ethene	Oxidation with H^+ / KMnO_4
D	Polymerise 1, 2-dichloroethene	Oxidation with H^+ / KMnO_4

[1 mark]

Question 4

A 1 mol sample of a monomer undergoes addition polymerisation and is completely polymerised.

How many moles of polymer might, theoretically, be formed?

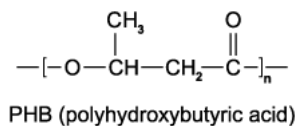
- 1 1
- 2 10^{-6}
- 3 $\frac{1}{6.02 \times 10^{23}}$

- A. 1 only
- B. 1 and 2
- C. 2 and 3
- D. 1, 2 and 3

[1 mark]

Question 5

Some micro-organisms naturally produce a polymer called PHB (polyhydroxy butyric acid). PHB is readily biodegradable.



What type of reaction will cause PHB to break down?

- A. Addition
- B. Reduction
- C. Hydrolysis
- D. Substitution

[1 mark]

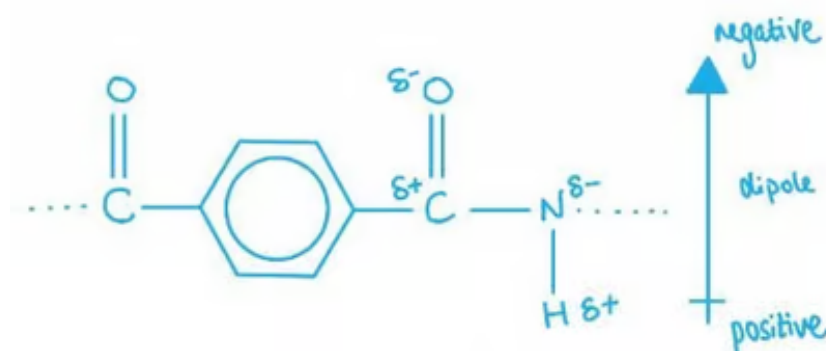
Model Answers: Hard

1

The correct answer is **C** because:

- (3) Steel is iron atoms mixed with chromium and carbon mixed, these atoms are much heavier than the carbon, nitrogen and oxygen found in *Kevlar*.
- (2) Kevlar aligns due to hydrogen bonding between δ^- oxygen and δ^+ hydrogen.
- Hydrogen bonds are stronger intermolecular forces than van der Waals forces that will hold 'strings' of steel rope together.
- Both these factors result in Kevlar being a lighter and stronger alternative to steel rope.

A, B & D are incorrect as *Kevlar* **does** have a **permanent dipole** because the oxygen in the carbonyl ($=O$) group is electronegative and will therefore be slightly negative (δ^-) and the hydrogen in the NH group will be slightly positive (δ^+) so the dipole will be positive to negative from top to bottom:



2

The correct answer is **C** because:

- In this question ***n*** is the number of **propene monomers** in the polymer molecule.
- Therefore, you must complete a balanced equation for the **complete combustion**

of one propene molecule and all stoichiometry (numbers of moles of each reactant and product) will then be **multiplied by n** .

- We know that complete combustion occurs due to the question stating 'burned in an **excess** of air'.
- The equation for the complete combustion of propene is:
 - $\text{C}_3\text{H}_6 + 4\frac{1}{2}\text{O}_2 \rightarrow 3\text{CO}_2 + 3\text{H}_2\text{O}$
- This means that on each side there are: **3** carbon, **6** hydrogen, **9** oxygen
 - Therefore, for n propene monomers the equation is:
 - $\text{C}_3\text{H}_6 + 4\frac{1}{2}n\text{O}_2 \rightarrow 3n\text{CO}_2 + 3n\text{H}_2\text{O}$

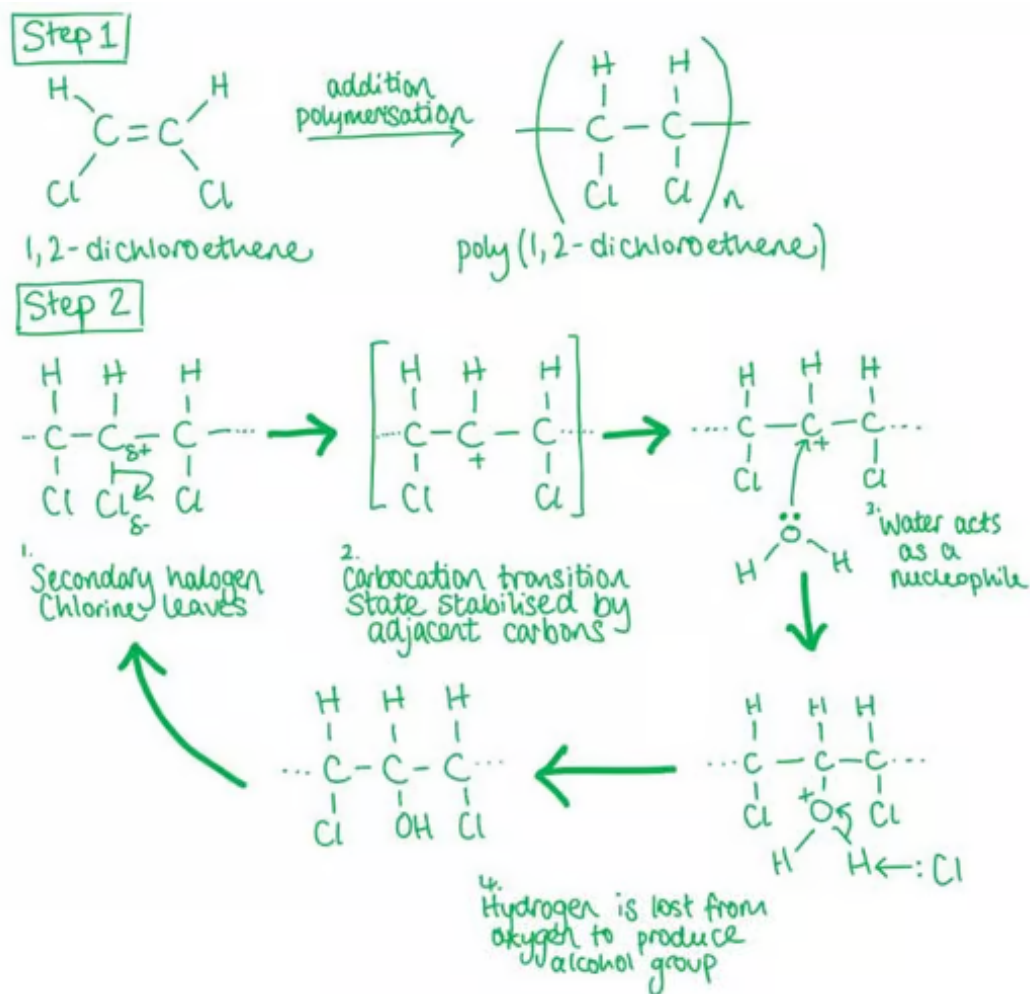
A is incorrect as poly(propene) is **burns easily**, it is 'highly flammable'.

B is incorrect as this burning occurs in an excess of air and therefore complete combustion occurs; carbon dioxide will be formed **not carbon**.

D is incorrect as the **stoichiometry** is incorrect.

3

The correct answer is **B** because:



A & C are incorrect as polymerising ethene forms repeating units with the structure of ethane – alkanes are unreactive and would not take part in further reaction to create an alcohol.

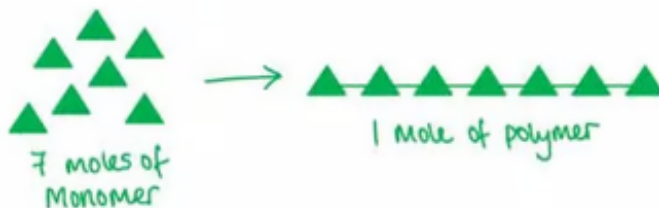
D is incorrect as **alcohol groups** are **not** produced from chlorine groups through oxidation with $\text{H}^+ / \text{KMnO}_4$.

4

The correct answer is **C** because:

- The number of moles of monomer **cannot** be the same as the number of moles of polymer.
- A polymer is a sequence of monomers **added together**.
- Therefore, the number of **moles of a polymer** must be **less** than the **moles of monomers** as multiple monomers make a polymer.

- For example:



Statement 1 is incorrect as the number of moles of monomer cannot be the same as the number of moles of polymer – therefore both cannot be 1.

5

The correct answer is **C** because:

- This polymer is made of an **ester repeating unit** as you can see from the groups – O–C– at the beginning and C=O at the end.
- If the monomer has an alcohol on the first carbon and a carboxylic acid on the last carbon, they can add together to make a polymer.
- Esters bonds are broken in hydrolysis reactions.
- 'Hydro' – with water, 'lysis' – to cut/break.

